FR

Fee Reconciliation Desk

FEE PAYMENT MATCHING AND RECONCILIATION

RELEASE NOTES V1.0

Release Notes v1.0

What is in the first release, verified results and known limitations

Document ID FRD-RN-07 · **Version** 1.0 · **Date** 27-Oct-2026

Prepared by Yasin Jariwala

Audience Everyone installing or evaluating Fee Reconciliation Desk

Document control

Document	Release Notes v1.0
Document ID	FRD-RN-07
Version	1.0
Date	27-Oct-2026
Prepared by	Yasin Jariwala
Product	Fee Reconciliation Desk v1.0
Audience	Everyone installing or evaluating Fee Reconciliation Desk
Classification	Public
Source	github.com/yasin679/FeeReconciliationApp · docs/release-notes.md

Version history

Version	Date	Author	Changes
1.0	27-Oct-2026	Yasin Jariwala	First release

Related documents

ID	Document
FRD-UG-01	User Guide
FRD-IG-02	Installation and Administration Guide
FRD-QS-03	Quick Start
FRD-FAQ-04	FAQ and Troubleshooting
FRD-TD-05	Technical Design
FRD-UAT-06	UAT Test Cases

Contents

1 Release summary 3

2 What is in v1.0 3

2.1 Bank statement import 3

2.2 Matching 3

2.3 Unmatched payments queue 3

2.4 Allocation and maker-checker approval 4

2.5 Idempotent posting and receipts 4

2.6 Reversals 4

2.7 Corrections 4

2.8 Cash 4

2.9 Student portal 4

2.10 Administration, audit and reporting 4

3 Security in v1.0 5

4 Verified results 5

4.1 Fixed during release testing 5

5 Known issues and limitations 6

6 Installation notes 6

7 Roadmap 6

1 Release summary

Item	Value
Product	Fee Reconciliation Desk
Version	1.0 (first release)
Date	27-Oct-2026
Platform	ASP.NET Core 8 (C#). Windows 10/11, Linux or macOS. No database server, no third-party packages.
Package	`Fee-Reconciliation-Desk.zip` (ready-built), or the source in the Git repository
Also included	A single-file web preview with sample data and a role switcher, for evaluation without installation

Version 1.0 matches every fee payment to the right student and fee component. A second person approves each allocation, and each payment is posted once with one receipt per line. Every rupee is accounted for in a daily books check.

2 What is in v1.0

2.1 Bank statement import

Finance imports the CSV exported from the college bank account. The import is the only way bank money enters the system.

- **Duplicates:** checked by source and transaction ID, ignoring case, spaces and punctuation. Re-importing a file is safe.
- **Same ID, different amount:** held as a conflict and never counted.
- **Rejected rows:** a missing ID, text in the amount, a zero amount or more than ₹10 crore. Rejected rows are counted in the result line.
- **Bank column names:** common names such as `UTR`, `Narration`, `Remitter` and `Mobile` are recognised.

2.2 Matching

A valid student ID in the bank reference matches automatically (policy AM-1). Split remarks such as `S101 15000 / S104 15000` become allocation lines. Payer names and phone numbers only suggest candidates, graded High, Medium or Low with the reason shown. A payment that arrives before the student is registered is held and matched once registration arrives.

2.3 Unmatched payments queue

One row per payment needing a person, showing:

- Verification state
- Candidates

- Status
- Next action, owner and age

Stage tiles and filter chips narrow the list, and a search finds a payment by ID, UTR, payer or student.

2.4 Allocation and maker-checker approval

Lines are entered by student and fee component and can never total more than was received minus reversals. A name-only match needs recorded evidence. A different Finance approver must approve, and the free funds are checked again at approval. An overpayment needs an approval note.

2.5 Idempotent posting and receipts

Each posting carries a unique key, so a retry after a student-system timeout writes exactly once. One receipt is issued per posted line, and receipt numbers never change.

2.6 Reversals

Bank reversals are linked to the original payment, capped at what remains, and applied by a Finance approver. The original keeps its amount and the receipt shows the adjustment. A reversal that cannot be applied becomes an exception, which can be re-linked or closed with a note.

2.7 Corrections

A correction is requested by one person and approved by a different Finance approver. It writes a compensating entry, keeps the original line visible, cancels the receipt with the reason, and frees the money to be allocated again.

2.8 Cash

Counsellors report cash and attach the receipt-book and deposit-slip numbers. Finance verifies the deposit; the person who reported it never can. Cash stays out of every balance until it is verified.

2.9 Student portal

Students see:

- Their own fee ledger and receipts
- Payments still being checked

They can report a payment with its UTR and an optional screenshot, and raise mismatch cases. A claim never reveals whether a matching bank line exists, and each student can have at most five claims under review.

2.10 Administration, audit and reporting

- **Administration:** users and roles; policy controls (AM-1, AA-1, the claim limit); a built-in system check.
- **Audit log:** append-only and hash-chained, with an integrity check.
- **Reporting:** a reconciliation CSV export, and a books check on the Overview (verified = posted + awaiting approval + unallocated, control difference ₹0).

3 Security in v1.0

Area	What is in place
Sign-in	PBKDF2-SHA256 password hashes (210,000 iterations); lockout after 5 failures; rate limit per address; no user-name enumeration by message or timing
Sessions	HttpOnly, SameSite=Strict cookies; idle timeout; sessions end on role change, disable or password reset
Requests	Antiforgery token on every change; API rate limit; body size limits
Headers	Strict Content Security Policy; frames, sniffing and referrers blocked; no external resources
Data	Role-filtered snapshots; students see only their own records; CSV export neutralises formulas
Storage	One lock; atomic save with rolling backup; damaged file stops start-up instead of being overwritten; session keys protected with DPAPI on Windows

4 Verified results

Every suite below passed on the release build on 27-Oct-2026.

Suite	Scope	Result
Built-in system check (C#)	27 scenarios covering duplicates, splits, cash, screenshots, reversals, corrections, retries, stale edits, late registration, the two-person rule, input limits and audit tamper detection	27 / 27 pass
Browser end-to-end	14 journeys, signed in as each real role against the running server	14 / 14 pass, 0 page errors
API attack suite	19 checks, including 11 role-escalation attempts	0 findings open
Concurrency	20 simultaneous proposals and parallel approvals on one ₹25,000 payment	1 proposal, 1 approval, 1 posting, 1 receipt
Sign-in rate limit	35 rapid attempts from one address	Attempts 31 onwards refused
Random actions	24,000 random actions by every role on the rules engine	No invariant violations
Web preview	27 live scenarios, injection and data-leak checks, network capture	All pass; the page makes no network requests

4.1 Fixed during release testing

Issue	Fix
A wrong password was reported as "server could not be reached"	Sign-in errors are now shown as sign-in errors
Adding a user lost the typed input when the list refreshed	Only the list refreshes

Issue	Fix
Sign-in time revealed which user names exist	Unknown users now cost the same as wrong passwords
A malformed request returned a server error	Treated as an empty request and refused cleanly
A burst of sign-ins could stall other users	Password checks run outside the shared lock
The sign-in limit was too strict for a shared office address	Made configurable (`LoginAttemptsPerMinute`, default 30)

5 Known issues and limitations

Limitation	Workaround or plan
Single machine and single data file	Host on one server behind HTTPS and back up `app/data` daily
No live bank or student-system connection	Bank money arrives by CSV import; students by manual entry. A bank API, virtual accounts per student and a student-system connector are planned.
Refund payouts are out of scope	Reversals record money the bank returned; refunds remain a Finance process
Policy changes need only one administrator	Every change is audited; a Finance sign-off step is planned
Receipts are on screen only	A PDF receipt with the institution's letterhead is planned
Fee amounts are fixed per component	Admission ₹25,000 and Tuition ₹1,20,000 in the sample data; per-student amounts from the student system are planned
Not yet run on Windows by a user	Built and tested on Linux with the same .NET 8 runtime. Run the system check and one sample-data session first.

6 Installation notes

1. Install the ASP.NET Core 8 runtime: `winget install Microsoft.DotNet.AspNetCore.8`.
2. Unzip `Fee-Reconciliation-Desk.zip` and run `start.bat`. From the Git source, run `build.bat` first; this needs the .NET 8 SDK.
3. Set `EnableDemoData` to false before the first real start.

Full steps are in the Installation and Administration Guide (FRD-IG-02).

7 Roadmap

Item	Benefit
Virtual account number per student	Removes name-only matching at the source

Item	Benefit
Bank API feed	Same-day verification, no CSV handling
Student-system posting connector	Ledger of record stays in the student system
Finance sign-off for policy changes	Two-person control on policy too
PDF receipts	Official receipts students can download
Per-student fee schedules	Instalments and concessions